

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B.Tech Examination (CE)

May 2012

CE209: Design & Analysis of Algorithm

Date: 18.05.2012, Friday

Time: 01:30 p.m to 4:30 p.m

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

- (a) Explain direct and indirect criteria to judge an algorithm with suitable example. 2
- (b) What is the best case complexity of quick sort and outline why it is so. How could its worst case behavior arise? 2
- (c) What are the general characteristics of Greedy Algorithms? Explain with suitable example. 3

Q-2

- (a) Write General recurrence equation for divide and conquer technique. Explain how Strassen's Matrix Multiplication can be done efficiently by using divide and conquer technique? 6
- (b) Explain the limit rules for Asymptotic notations(O , Ω , θ) and give the relationship among them. 4
- (c) Define P, NP, NP complete and NP-Hard problems. 4

OR

Q-2

- (a) Write an algorithm to implement Depth-first search(DFS)? How is Depth-first search different from Breadth-first search? What are the applications of DFS? 6
- (b) Find the running time of given algorithm in terms of no of primitive operations required in best case, average case and worst case. 4

Algorithm prefix Averages(X, n)

Input array X of n integers

Output array A of prefix averages of X

$A \leftarrow$ new array of n integers

$s \leftarrow 0$

for $i \leftarrow 0$ to $n - 1$ do

$s \leftarrow s + X[i]$

$A[i] \leftarrow s / (i + 1)$

return A

- (c) Analysis of algorithm is performed with respect to a computational model called Random Access Machine (RAM). Explain the significance of RAM Model in analysis of algorithm. 4

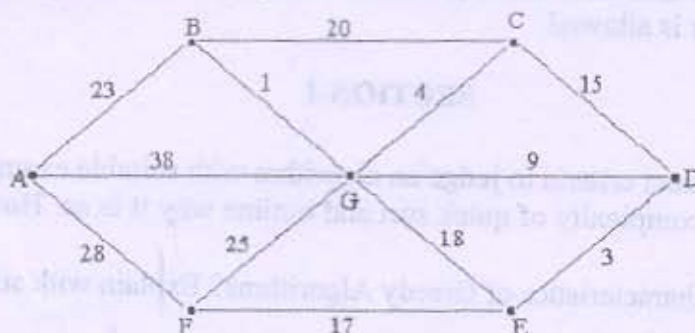
Q-3 Attempt Any Two.

- (a) Sort the following set using quick sort algorithm.
{12, 34, 45, 15, 4, 11, 7, 8, 5, 14, 35, 89, 43, 21}. Choose first element as a pivot element.
Show under what order of input, the quick sort will have worst-case and best case situations. 14

- (b) Let $S = \{1, 2, \dots, n\}$ is set of jobs and each job i has a deadline $d_i \geq 0$ and a profit $p_i \geq 0$. We need one unit of time to process each job and we can do at most one job each time. We can earn the profit p_i if job i is completed by its deadline. Write Greedy algorithm for Solving above job scheduling problem and solve the following problem using algorithm.

I	1	2	3	4	5
p_i	20	15	10	5	1
d_i	2	2	1	3	3

- (c) Consider the following undirected graph:



1. Find the adjacency list representation of the graph.
2. Find a minimum cost spanning tree by Kruskal's algorithm.
3. List the nodes of the graph by breadth first search starting from node A.

SECTION-II

- Q-4
- (a) Explain the terms:
- i) Articulation Point
 - ii) Optimal Substructure property
- (b) What is the smallest value of n such that an algorithm whose running time is $50n^3$ runs faster than an algorithm whose running time is 3^n on the same machine? Justify your answer.
- (c) Give similarities and differences between Branch & Bound and Backtracking.
- Q-5
- (a) How the assembly line scheduling can be solved using dynamic programming? Explain with suitable example.
- (b) What is Finite Automata? How it can be used in string matching. Explain with the help of a suitable example.
- (c) Solve the following recurrence equation.
- $$T(n) = 4T(n/2) + n \log n \quad \text{otherwise}$$
- $$= 4 \quad \text{if } n=1$$

OR

- Q-5
- (a) What do you mean by complexity of an algorithm? Derive the asymptotic time complexity of a recursive and non recursive version of binary search algorithm.
- (b) Solve the following 0/1 Knapsack Problem using Dynamic Programming. There are five items whose weights and values are given in following arrays.
 Weight $w[] = \{1, 4, 5, 6, 7\}$ Value $v[] = \{1, 6, 18, 22, 28\}$
 Show your equation and find out the optimal knapsack items for weight capacity of 11 units.

- (c) Solve the following recurrence equation.

$$T_n = 2T_{n-1} + n, \quad \text{otherwise} \\ = 1 \quad \text{if } n=0$$

4

Q-6 Attempt Any Two.

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- (a) Consider the chain of matrices A1,A2,...,A6 with the dimensions given below. Give the optimal parenthesization to get the product A1..A6.

Matrix Dimension

A1: 30 X 35

A2 :35 X 15

A3 :15 X 5

A4 :5 X 10

A5 :10 X 20

A6 : 20 X 25

- (b) What is Backtracking ? Solve the following 0/1 Knapsack Problem using Backtracking.

There are five items whose weights and values are given in following arrays.

Weight $w[] = \{ 1,3,5,7,8 \}$ Value $v[] = \{ 2,9,15, 20, 23 \}$

Find out the optimal knapsack items for weight capacity of 8 units.

- (c) Solve the following using Recurrence Tree Method.

1. $T(n) = 3T(n/2) + n$

2. $T(n) = 2T(n/3) + n^2$
